

### Usability and Ergonomics:

The Frogger game is very simple to understand and operate on the DE1\_SoC FPGA Board. The buttons to press on the board to navigate your frog similarly represent an arcade machine, where the up and down movements are controlled by the left 2 keys, and the left and right movements controlled by the right 2 keys. The Switch reset button is out of the way so you won't accidentally hit it when pressing, and the 16x16 Bi-Color LED Board is bright enough so you can clearly distinguish among colors. Creating this simple design means that it doesn't take any technology knowledge or experience to operate.

### Sustainability:

The cars move slowly making it relatively easy for players to win, and whether a player wins or loses is clearly displayed on both the LED Board and the HEX Display. This game is also very fun to play!

### Cost and Resource Usage:

The highest Resource Utilization by Entity that we had before this lab was 108, but as a result of this final project the Resource Utilization increased to 966, which is pretty drastic. This means that running this lab requires the FPGA to run "more" than previously to support this lab, which could be due to both the user input and the fact that there's a whole new attachment in the 16x16 LED Board.

### Public Health & Safety, Welfare:

This poses no health or safety risks as an entertainment device. Having arcade games like Frogger can be thought of as nice stress relievers at the end of a long day, as a way to help you rewind.